M339D: March 23rd, 2026.

Binomial Option Pricing [cont'd].

Review.

PAYOFF

No arbitrage condition

$$d < e^{rh} < u$$

$$S(0) \begin{cases} S_u = u \cdot S(0) & V_u = v(S_u) \\ S_d = d \cdot S(0) & V_d = v(S_d) \end{cases}$$

$$\underbrace{}_{h=T}$$

$$\left.\begin{aligned} \Delta &= \frac{V_u - V_d}{S_u - S_d} \\ B &= e^{-rh} \cdot \frac{u \cdot V_d - d \cdot V_u}{u-d} \end{aligned}\right\} \text{Replicating Portfolio}$$

$$\boxed{V(0) = \Delta \cdot S(0) + B}$$

.... algebra ....

$$\boxed{\begin{aligned} V(0) &= e^{-rT} \mathbb{E}^*[V(T)] \\ &= e^{-rT}\left[p^* \cdot V_u + (1-p^*) \cdot V_d\right] \\ \text{w/} \quad & \boxed{p^* = \frac{e^{rh} - d}{u-d}} \end{aligned}}$$

Special Case: Forward Binomial Tree.

$$\boxed{\sigma \text{ ... volatility}}$$

$$\boxed{\begin{aligned} u &:= e^{rh + \sigma\sqrt{h}} \\ d &:= e^{rh - \sigma\sqrt{h}} \end{aligned}} \quad \longleftarrow$$

The risk-neutral probability:

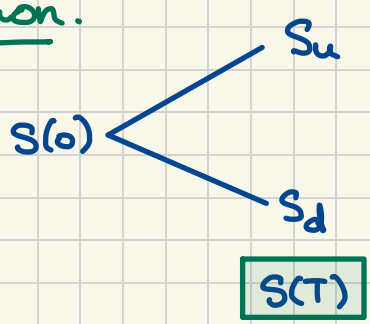
$$p^* = \frac{e^{rh} - d}{u-d} = \frac{\overset{1}{\cancel{e^{rh}}} - e^{\cancel{rh} - \sigma\sqrt{h}}}{e^{\cancel{rh} + \sigma\sqrt{h}} - e^{\cancel{rh} - \sigma\sqrt{h}}}$$

$$p^* = \frac{1 - e^{-\sigma\sqrt{h}}}{e^{\sigma\sqrt{h}} - e^{-\sigma\sqrt{h}}} \cdot \frac{e^{\sigma\sqrt{h}}}{e^{\sigma\sqrt{h}}} = \frac{\overset{1}{\cancel{e^{\sigma\sqrt{h}} - 1}}}{\underbrace{e^{2\sigma\sqrt{h}} - 1}_{\cancel{(e^{\sigma\sqrt{h}} - 1)}(e^{\sigma\sqrt{h}} + 1)}}$$

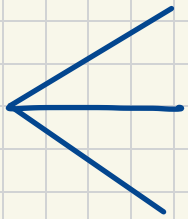
$$\boxed{p^* = \frac{1}{1+e^{\sigma\sqrt{h}}}} \xrightarrow[h \to 0]{} \frac{1}{2}$$

The shortcut $\boxed{\text{ONLY}}$ for the forward binomial tree.

<u>Motivation.</u>



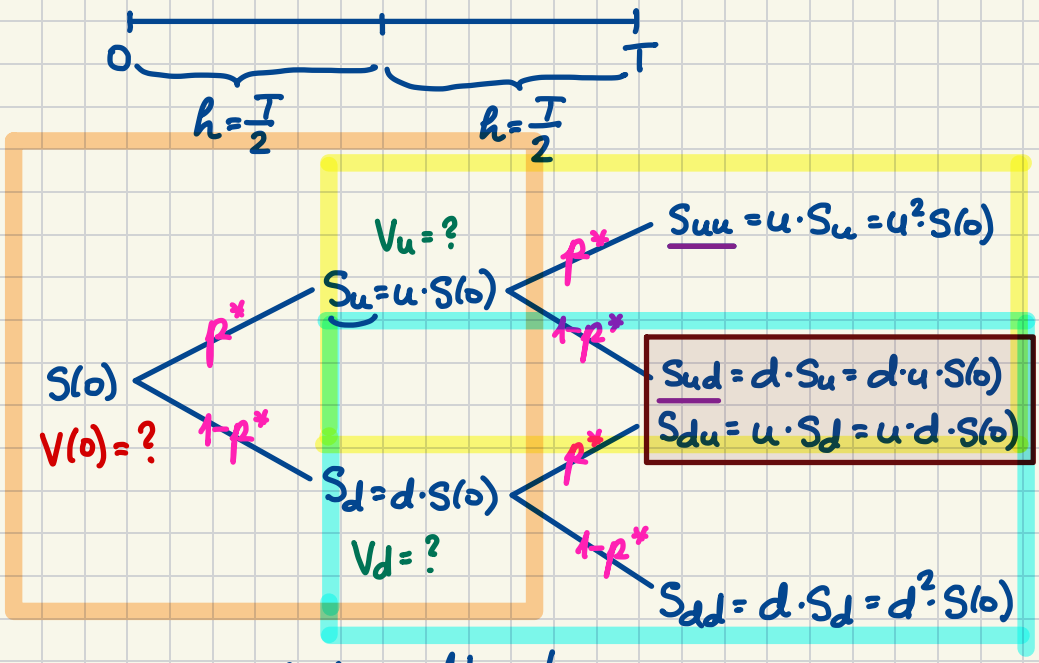
Q: How can we make the model for $S(T)$ richer, but still "<u>interpretable</u>"!



<u>k-nary tree</u>

Two Periods.

$\boxed{n=2}$



Payoff

$V_{uu}=v(S_{uu})$

$V_{ud}=v(S_{ud})$

$V_{dd}=v(S_{dd})$

populating the tree →

← pricing the option

- up node: replicating portfolio for the option

$$\begin{cases} \Delta_u = \dfrac{V_{uu}-V_{ud}}{S_{uu}-S_{ud}} \\ B_u = e^{-rh}\cdot\dfrac{u\cdot V_{ud}-d\cdot V_{uu}}{u-d} \end{cases}$$

$\Rightarrow$ the option's value in the up node:

$$V_u=\Delta_u\cdot S_u+B_u=e^{-rh}\cdot\left[p^*\cdot V_{uu}+(1-p^*)\cdot V_{ud}\right]$$

w/ $\boxed{p^*=\dfrac{e^{rh}-d}{u-d}}$

- down node: Δ_d, B_d

$$\Rightarrow \quad V_d=\Delta_d\cdot S_d+B_d=e^{-rh}\cdot\left[p^*\cdot V_{ud}+(1-p^*)\cdot V_{dd}\right]$$

w/ $p^*=\dfrac{e^{rh}-d}{u-d}$

- **ROOT node:**

$$\Delta_0 = \frac{V_u - V_d}{S_u - S_d}$$

$$B_0 = e^{-rh} \cdot \frac{u \cdot V_d - d \cdot V_u}{u - d}$$

$$\Rightarrow \boxed{V(0) = \Delta_0 \cdot S(0) + B_0}$$

From the "risk-neutral perspective":

$$V(0) = e^{-rh} \cdot \left[p^* \cdot V_u + (1-p^*) \cdot V_d \right]$$

$$= e^{-rh} \cdot \left[p^* e^{-rh} \left(p^* \cdot V_{uu} + (1-p^*) \cdot V_{ud} \right) + \right.$$

$$\left. + (1-p^*) e^{-rh} \left(p^* \cdot V_{ud} + (1-p^*) \cdot V_{dd} \right) \right]$$

$$= e^{-r(2h)} \underbrace{\left[(p^*)^2 \cdot V_{uu} + 2 \cdot p^*(1-p^*) \cdot V_{ud} + (1-p^*)^2 \cdot V_{dd} \right]}_{\text{Risk-Neutral Expectation of the Payoff}}$$

Generally:

$$\boxed{V(0) = e^{-rT} \, \mathbb{E}^*[V(T)]}$$